

The Hidden Data Science Platform In SAP HANA



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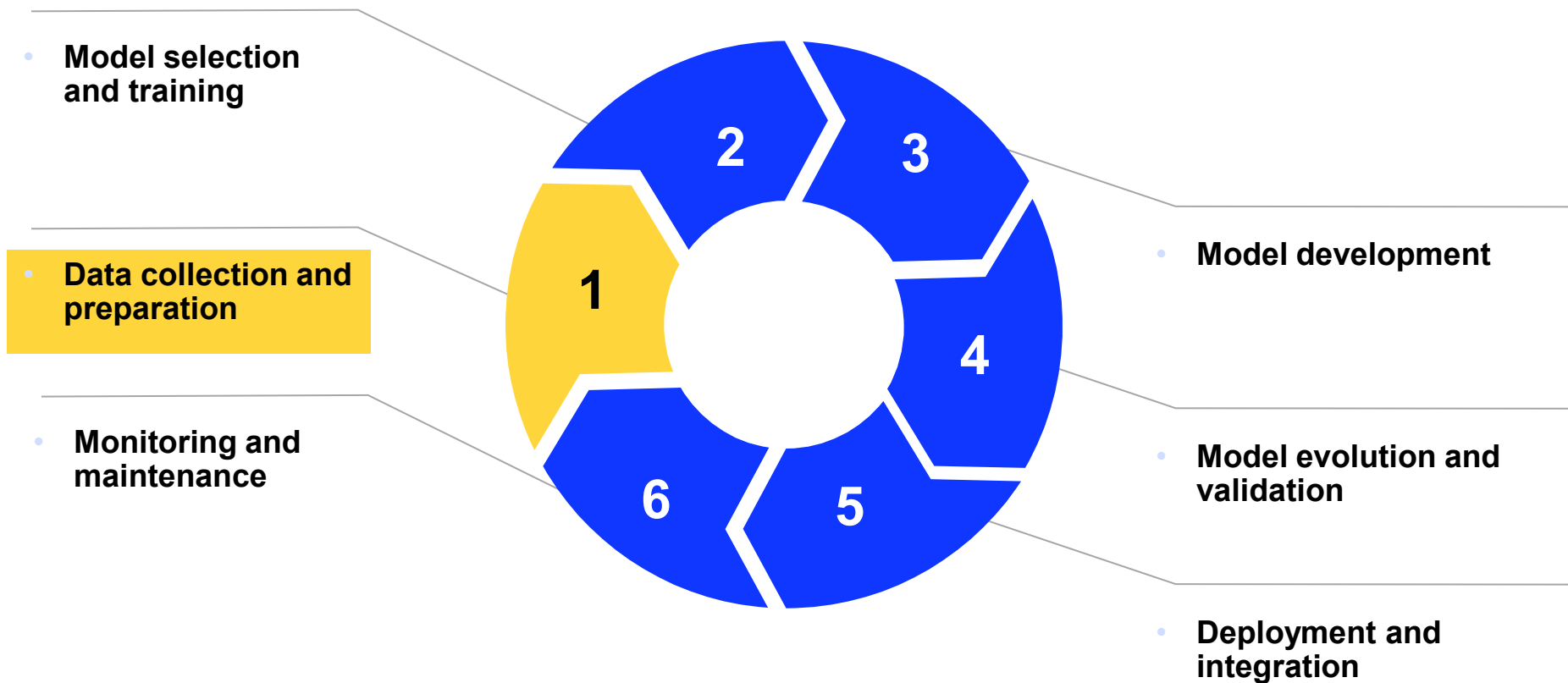
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Houston , We have a situation!

Every AI project starts with data.





Data Movement Is an Architecture Choice



hana_ml | Python for In-Database Machine Learning

- **hana-ml** is SAP's **Python client** for building and running machine learning directly on SAP HANA.

- **SAP HANA DataFrames**
Python interface acts as a reference to the data, rather than copying it into local memory

- **Native Machine Learning APIs** : Feeds remote data references through directly into HANA's built-in PAL & APL for high-speed model training.

```
from hana_ml import dataframe

# 1. Establish connection to the SAP HANA instance
conn = dataframe.ConnectionContext(
    address="your_hana_host",
    port=30015,
    user="your_username",
    password="your_password"
)

# 2. Create a HANA DataFrame (Data stays in the DB)
hana_df = conn.table('SALES_DATA', schema='BI_SCHEMA')

# 3. Apply remote filtering
filtered_df = hana_df.filter("REGION = 'EMEA'").select('PRODUCT_ID', 'REVENUE')

# 4. Optional: Explicitly pull data to local client as Pandas
local_pandas_df = filtered_df.collect()
```

hana_ml | From HANA Cloud to Business Data Cloud

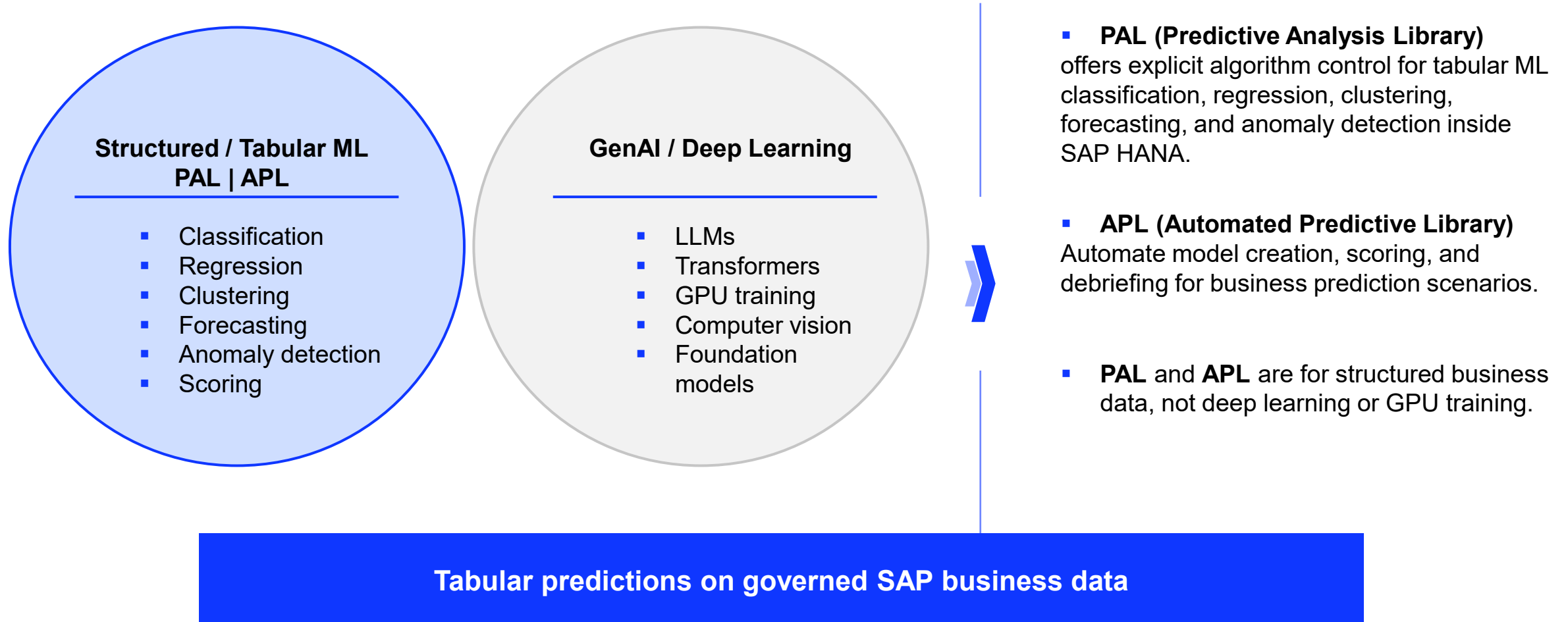
- **Remote Push-Down Execution:**
Automatically translates Python DataFrame manipulations into native SQL, processing massive BDC datasets entirely in-database without downloading data locally.
- **In-Database AutoML Pipelines:**
Orchestrates advanced ML tasks natively within BDC
- **Model Persistence & BI Expose:**
Saves trained model artifacts directly back to the cloud layer, enabling batch inference results to be wrapped into live Open SQL views for SAP Analytics Cloud.

```
import hana_ml.dataframe as dataframe

# Connect to the underlying cloud database engine behind
# Datasphere/BDC
conn = dataframe.ConnectionContext(
    address="://ondemand.com",
    port=443,
    user="MY_SPACE#BI_USER",          # Technical Space user
    created in tenant
    password="SecureDatabasePassword",
    encrypt=True,                    # Enforce TLS encryption
    for cloud nodes
    sslValidateCertificate=True
)

print("Connection status:", conn.connection.isconnected())
```

Predictive ML | PAL and APL as In-Database Predictive Engines



SAP HANA Cloud Vector & Knowledge Graph Engines| One Query, Three Engines

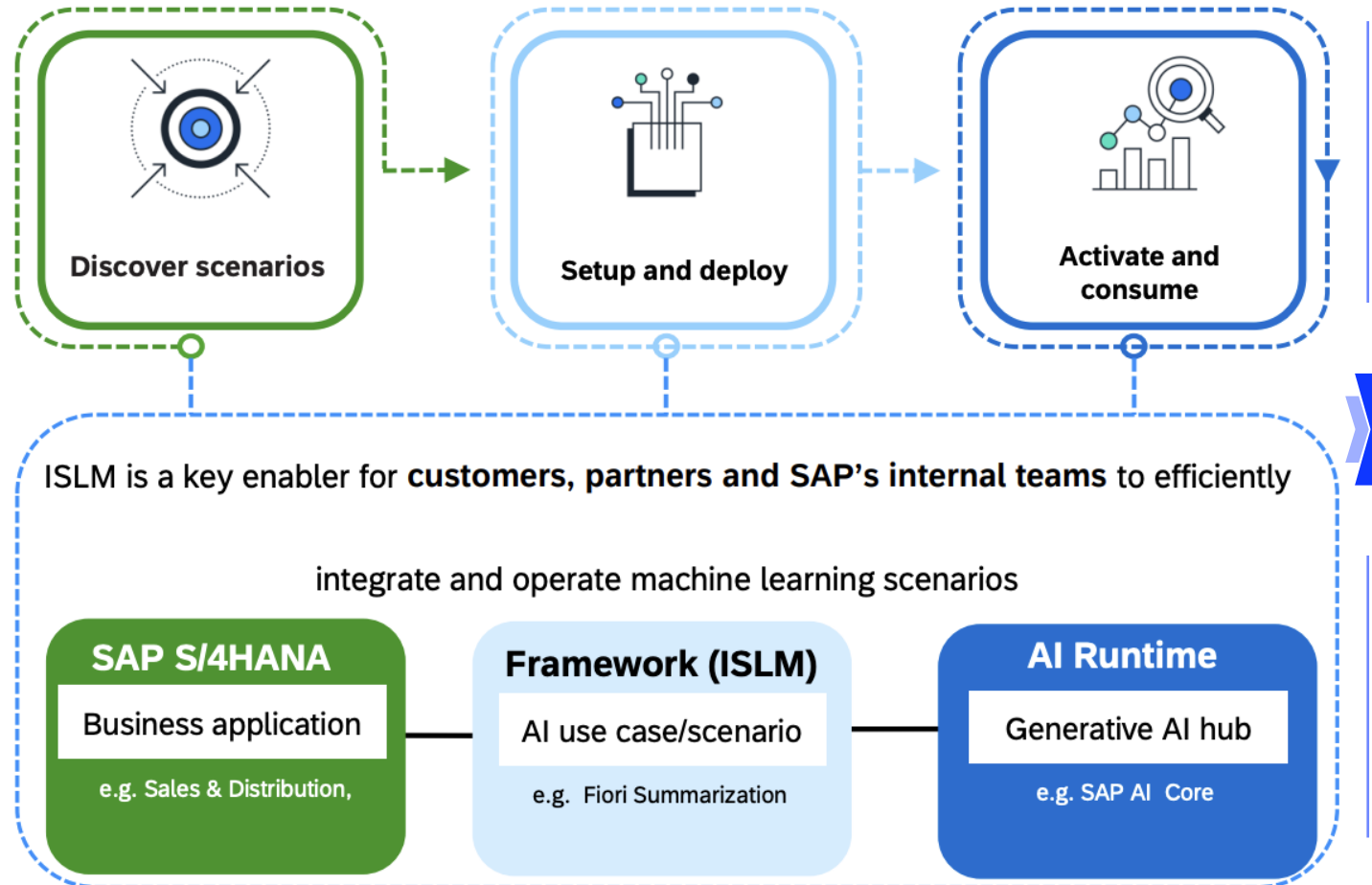
- **Knowledge Graph Engine:**
Connects business entities and their relationships.
- **Relational Engine:**
Stores and queries structured business data in tables.
- **Vector Engine:**
Finds meaning-based similarity across text, documents, or embeddings.



```
CALL SPARQL_EXECUTE('
  SELECT ?supplier ?riskNote WHERE {
    GRAPH <kg_suppliers> {
      ?supplier a <Supplier> ;
                <hasRegion> "EMEA" .
    }

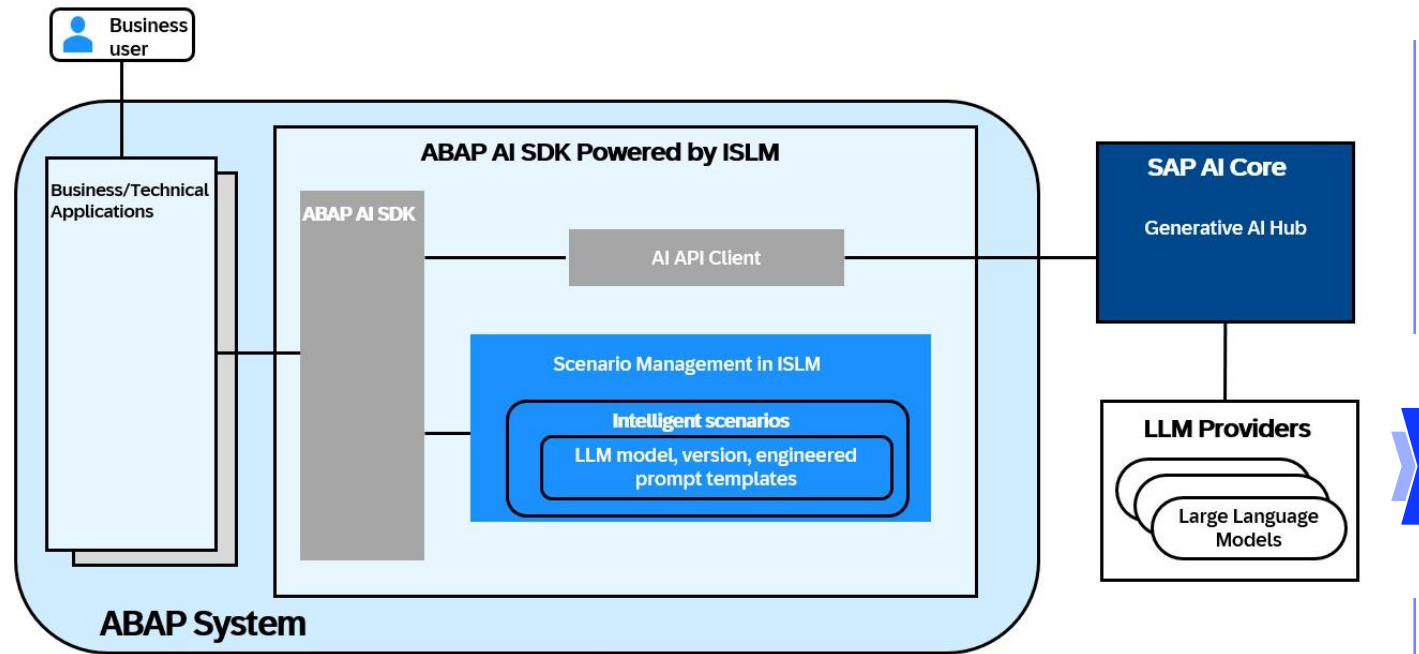
    SQL_TABLE("
      SELECT
        SUPPLIER_URI AS supplier,
        NOTE_TEXT AS riskNote
      FROM SUPPLIER_RISK_NOTES
      WHERE COSINE_SIMILARITY(
        NOTE_EMBEDDING,
        VECTOR_EMBEDDING(''customs delay'',
        ''QUERY'', ''SAP_NEB.20240715'')
        ) > 0.75
      ")
    }', '?', '?', '?');
```


ISLM | The operational bridge from HANA ML to business action



- **From model to managed scenario**
ISLM turns a trained model into a registered, versioned, and consumable business scenario.
- **Embedded when the signal lives in HANA**
Use PAL/APL models close to governed SAP data and operational processes.
- **Side-by-side when the workload demands it**
Use SAP AI Core or BTP services for external runtimes, orchestration, or LLM-based extensions.

ISLM | Scaling Generative AI in S/4HANA



- **Integration with the Orchestration Service**
ISLM now integrates with the Orchestration Service to deliver advanced, enterprise-grade capabilities for Generative AI scenarios
- **Support for new models offered by Generative AI Hub:**
leverage state-of-the-art large language models while maintaining flexibility and future readiness.
- **Simplification of creation of Intelligent Scenario**
moving from a pro-code approach to a more streamlined low-code experience and allows the optional use of the prediction class for Generative AI scenarios.

Payment Behavior Index (PBI) | categorize customers by payment default risk and is the backbone for the flexible Dunning Process

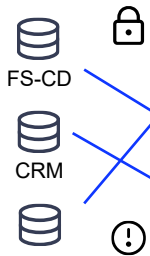
Illustrative

From data

Fragmented payment data in multiple systems – no insights on customers

Manually extracting and classifying information from siloed payment data

Accessing data



Processing data across systems

Customer Ratings from CRM
Dunning defaults
Payment information

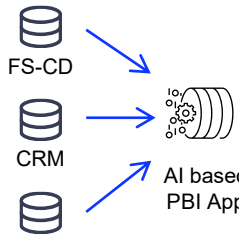
- Payment data is scattered in multiple systems, without sufficient links
- Unable to derive insights in customer's payment behavior

To insights

Harnessing power of data to generate AI-based Payment Behavior Index

AI-powered PBI application delivers real-time Payment Behavior Index

Accessing data



Processing PBI across systems

Customer categorization by risk
Personalized dunning and collections
Personalized communication
Using **insights** to steer dunning

- Dynamic score summarizing customer payment behavior using AI
- Analyzes large volumes of payment and behavioral data from various systems
- Enabling far deeper insights than traditional rule-based scoring

Payment Behavior Index App

Key benefits

- **Customer Centricity and Operational efficiency** enabling smarter, real-time collections actions
- **Responsible AI** ensuring data privacy with robust anonymization and compliance, following SAP's ethical AI framework
- **Future-proof solution** for 's needs, e.g., support of any additional parameters, AI development is in line with SAP's Business AI roadmap



Take away | Place intelligence where it structurally belongs



Run Close To HANA

When Signal, Context and governance live there



Govern through ISLM

When prediction is required



Integrate with BTP / AI Core

When the workload needs orchestration, LLMs, or GPU

Q&A Thank you

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Mentor 🧡 #womenInTech 🖥️



SAP Fioneer 